Project Usage Instructions

Overview

The project is split across two GitHub repositories:

- ➔ OR-Recorder-Transcriber
- ➔ NOL-Event-Data-Classifier

1. Install Prerequisites

Install Python 3.12+ (tested with 3.12.10)

Microsoft Store - Python 3.12

or

python.org - Python 3.12.10 release

(required on Linux)

2. Download OR-Recorder-Transcriber

Go to the project's GitHub page:

OR-Recorder-Transcriber

then open the releases page:

OR-Recorder-Transcriber releases

Download the latest executable

- ORRT.exe for Windows
- For Linux, go to the SourceForge project page and download the ORRT executable:
  OR-Recorder-Transcriber on SourceForge

Get the source code

Clone the project with Git:

```
git clone https://github.com/Gabrieleirbag1/OR-Recorder-Transcriber.git
```

or download the source code as a ZIP archive:

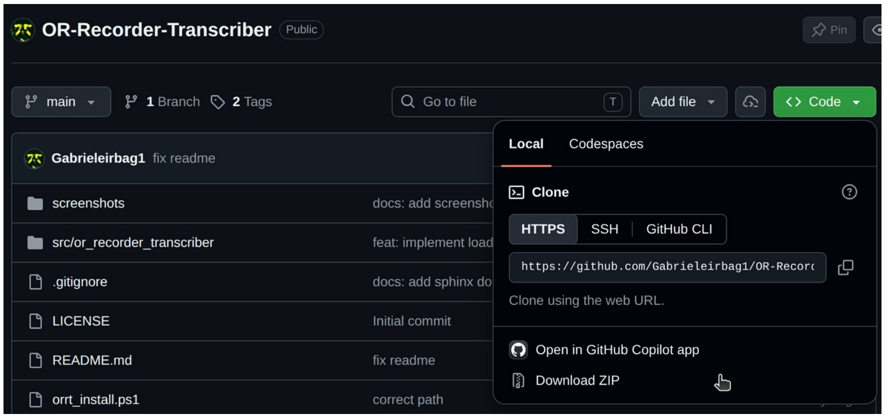

Figure 1 - Cloning or downloading the repository from GitHub

Follow the instructions in the README to install from source or run the Python code directly.

Note: The first time you run the executable, it may take a few minutes as it needs to download the audio transcription and language models. Don't worry, this is completely normal if the application is not opening at first!

3. Fine-Tune the Models for Your Labels

Get the classifier project:

[NOL-Event-Data-Classifier]

Install it by following the README.

Set your custom training labels

To use your own labels for training, edit the following file and replace the label list with your own:

`src/nol_event_classifier/supervised/labels.py`

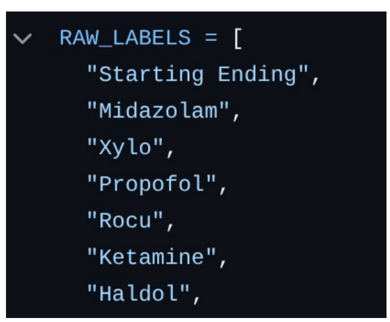

Figure 2 - labels.py

Run a sample fine-tuning job

```
# Fine Tuning (Supervised)
# -m: Base model name for SetFit
# -o: Output name
# -t: Test size for the train/test split
# -e: Number of epochs
# -b: Batch size
# -i: Number of iterations
python3 src/nol_event_classifier/supervised/fine_tuning.py \
  -m paraphrase-multilingual-MiniLM-L12-v2 \
  -o "setfit_paraphrase-multilingual-mpnet-base-v2_3ep-16bs-10it" \
  -t 0.2 \
  -e 3 \
  -b 16 \
  -i 10
```

This produces a trained model that you can then import into the other application.

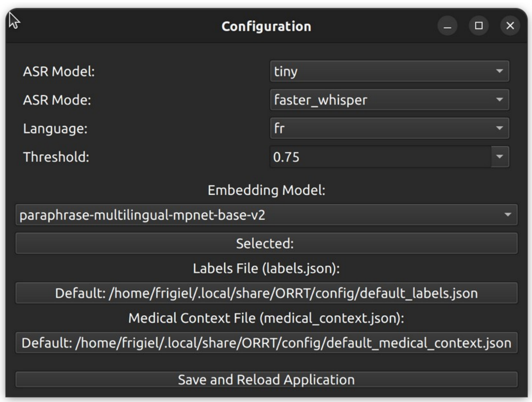

Figure 3 - OR-Recorder-Transcriber configuration window

4. Configure OR-Recorder-Transcriber

In the Configuration tab of OR-Recorder-Transcriber you can:

- Import a folder containing all your models, which you can then select - useful for models you've previously trained on your own labels.
- Provide the list of custom labels. Include either “Other” or “Medication” so that charts can be generated later.

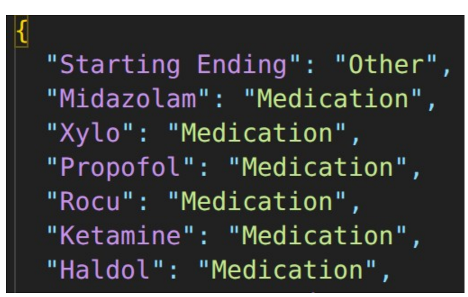

Figure 4 - Model and label configuration

- Set the medical context file the same way; it steers classification toward commonly searched terms - by default, terms from the field of operative anesthesia.

```
[
    "midazolam",
    "vs",
    "End",
    "cannula",
    "CGR",
    "Infiltration",
    "lithotomie",
    "trocart",
    "ml",
    "pr",
    "bolus",
```

Figure 5 - Medical context configuration

You can also select:

- The model size (tiny, base, small, medium, large)
- The library used (whisper, faster-whisper, pywhispercpp)
- The language (fr, en...)
- The confidence threshold that determines when the program will prompt you to manually re-select the labels chosen during classification